Framing

Definition: Framing is the process of dividing a continuous stream of bits received from the Network Layer into **smaller units called frames** before transmitting them over the physical medium.

The **Data Link Layer** performs framing by adding control information such as headers and trailers to the data.

Why is Framing Needed? When data is transmitted over a network, it is sent as a continuous stream of bits. The receiver must know:

- Where a frame starts.
- Where a frame ends.
- Which bits belong to one frame.

Framing provides clear boundaries so that the receiver can correctly identify and process each frame.

Functions of Framing

- Divides data into manageable frames.
- Identifies the beginning and end of each frame.
- Helps in error detection and flow control.
- Enables reliable communication between sender and receiver.

Example of Framing:

Suppose the sender wants to send: `HELLO`

The Data Link Layer adds a header and trailer.

```
+--------+-------+---------+
| Header | HELLO | Trailer |
+--------+-------+---------+
```
This complete unit is called a **Frame**.

2. Framing Methods

The Data Link Layer uses different techniques to identify frame boundaries.

Types of Framing

1. Character Count
2. Byte (Character) Stuffing
3. Bit Stuffing

4. Physical Layer Coding Violations

Bit Stuffing:

Definition: Bit Stuffing is a framing technique in which the sender inserts an extra **0** after every sequence of **five consecutive 1's** in the data.

This prevents the receiver from confusing the data with the **flag pattern** used to mark the beginning and end of a frame.

Why is Bit Stuffing Required? In protocols such as **HDLC**, the flag used to indicate frame boundaries is: `01111110`

If the same pattern appears inside the transmitted data, the receiver may incorrectly assume that the frame has ended.

To avoid this ambiguity, a **0** is inserted after every five consecutive **1's**.

Example:

Original Data

`111111`

After Bit Stuffing

`1111101`

Here:

`11111`

After five consecutive **1's**, an extra **0** is inserted.

Frame Sent

`01111110 1111101 01111110`

where:

- First `01111110` → Starting Flag
- `1111101` → Stuffed Data
- Last `01111110` → Ending Flag

Advantages of Bit Stuffing

- Prevents flag confusion.
- Suitable for binary data.
- Used in HDLC and similar protocols.
- Ensures reliable frame transmission.

Byte (Character) Stuffing:

Definition

Byte Stuffing (also called Character Stuffing) is a framing technique in which an **escape character (ESC)** is inserted before any special control character that appears in the data.

This ensures that the receiver treats it as normal data rather than as a frame delimiter.

Why is Byte Stuffing Required?

Suppose the protocol uses:

```
FLAG
```

to indicate the start and end of a frame.

If the data itself contains:

```
FLAG
```

the receiver may incorrectly assume the frame has ended.

To prevent this, the sender inserts:

```
ESC
```

before the special character.

Example

Original Data

```
ABCFLAGXYZ
```

After Byte Stuffing

```
ABCESCFLAGXYZ
```

Complete Frame

```
FLAG ABCESCFLAGXYZ FLAG
```

where:

- First FLAG → Start of Frame
- ESC → Escape Character
- Last FLAG → End of Frame

Advantages of Byte Stuffing

- Eliminates ambiguity caused by special characters.
- Suitable for character-oriented protocols.
- Easy to implement.
- Provides reliable frame identification.